Design and HRI

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Contents

Robot design approaches

- Inside-out
  - This process of design starts from the inside and builds up to the outside- solving technical issues first and designing the robot's appearance and behavior to fit.
  - This common approach to robot building is also known as the “Frankenstein approach": we take whatever technology is available and put it together to get certain robotic functions.
  - Lack of consideration of the social context of use within the design process can lead to surprising effects on robot interaction

TurtleBot2

Robot design approaches

- Outside-in
  - The design process starts from the interaction that we expect the robot to be engaged in, which will determine its outside shape and behaviors.
  - Once the design has been settled upon, we work all the technology into it.
  - This form of robot design often requires incorporating expertise from several disciplines. For example-
    - designers might work on developing specific design concepts,
    - social scientists may perform exploratory studies to learn about the potential users and context of use,
    - engineers and computer scientists need to communicate with the designers to identify how specific design ideas can be realistically instantiated in working technology.

Mythical robots designed from the outside to the inside.

Design Considerations in HRI

- **Robot morphology and form**
  - Should form follow function or vice versa?
  - In HRI, form and function are inherently interconnected and thus cannot be considered separately.
  - The design space of robots is relatively large and considers questions regarding the
    - form,
    - function,
    - level of autonomy,
    - interaction modalities, and
    - how all those fit with particular users and contexts.
  - An important aspect of the design is figuring out how to make appropriate decisions about these various design aspects.

Figure 4.3 Robovie-MR2 (2010) is a humanoid robot controlled through a cell phone.

Figure 4.4 Zoomorphic and minimalistic robots: Muu (2001–2006), Keepon (2003–present) and Naked Invisible Guy.

Figure 4.5 Sociable Trash Box robots are an example of robjects—robotic objects with interaction capabilities. (Source: Michi Okada)

Design Considerations in HRI

- **Robot morphology and form**
  - Contemporary HRI designers have several different forms of robots to choose from
    - **Androids and humanoids** most closely resemble humans in appearance.
    - **Zoomorphic robots** are shaped like animals with which we are familiar (e.g., cats or dogs) or like animals that are familiar but that we do not typically interact with (e.g., dinosaurs or seals).
    - **Minimalist robots** explore the minimal requirements necessary for inspiring social HRI, such as Muu, or Keepon.
    - **Robjects** are interactive robotic artifacts whose design is based on objects rather than living creatures, e.g. social trashcans.

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Design Considerations in HRI

- **Affordance**
  - Describes the perceivable relationships between an organism and its environment that enable certain actions (e.g., a chair is something to sit on).
  - A designer needs to design a product while making its affordances explicit.
  - A robot's appearance is an important affordance because people tend to assume that the robot's capabilities will be commensurate with its appearance.
  - If a robot looks like a human, it is expected to act like a human; if it has eyes, it should see; if it has arms, it should be able to pick up things and might be able to shake hands.

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Design Considerations in HRI

- **Affordance: Key dimensions**
  - Physical affordances: These are the literal "action possibilities" provided by the robot's hardware. For example, a handle suggests it can be pulled, while a flat surface suggests something can be placed on it.
  - Social & Emotional Affordances: These communicate how a human should interact with the robot socially. A robot with big eyes might "afford" eye contact, while a soft, plush material invites "gentle touch".

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Design Considerations in HRI

- **Affordance: Key dimensions**
  - Cognitive Affordances: These are design elements that help users understand the robot's status or intentions. A progress bar or light that pulses slowly can "afford" the understanding that the robot is currently "thinking" or processing a task.
  - Functional Affordances: These define the robot's specific purpose. For example, a robot shaped like a courier clearly "affords" the delivery of items, whereas a humanoid might "afford" more complex collaborative tasks.

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Design Considerations in HRI

- **Affordance**
  - Think about the features of a robot tour guide in terms of “design affordances." Which affordances should be considered in humanlike robots?

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Design Considerations in HRI

- **Design patterns**
  - Patterns describe a problem that occurs over and over again in our environment, and then describe the core of the solution to that problem.
  - Patterns should be abstract enough that you can have several different instantiations, so that
    - they can be combined, so that less complex patterns can be integrated into more complex patterns and
    - they serve to describe interactions with the social and physical world.
  - For example, the didactic communication pattern (where the robot assumes the role of a teacher) could be combined with a motion pattern (where the robot initiates a movement and aligns it with the human counterpart of the interaction) to create a robotic tour guide.

Robotic tour guide

Design Considerations in HRI

- **Design patterns**
  - Try to think about “design patterns" for social robots that greet people daily. Find and describe repeatedly reused patterns in behavior.

Robotic tour guide

Design Considerations in HRI

- **Design principles:** Researchers have suggested following principles to consider when developing the appropriate robot forms, patterns, and affordances in HRI design.

  1. **Matching the form and function of the design:** If your robot is humanoid, people will expect it to do humanlike things- talk, think, and act like a human. If this is not necessary for its purpose, such as cleaning, it might be better to stick to less anthropomorphic designs.

  2. **Underpromise and overdeliver:** When people's expectations are raised by a robot's appearance or by introducing the robot as intelligent or companion-like, and those expectations are not met by its functionality, people are obviously disappointed and will negatively evaluate the robot.

Figure 2.6 Paro (2003–present), a social robot made to resemble a baby harp seal. Paro is provided as a social companion robot. (Source: National Institute of Advanced Industrial Science and Technology)

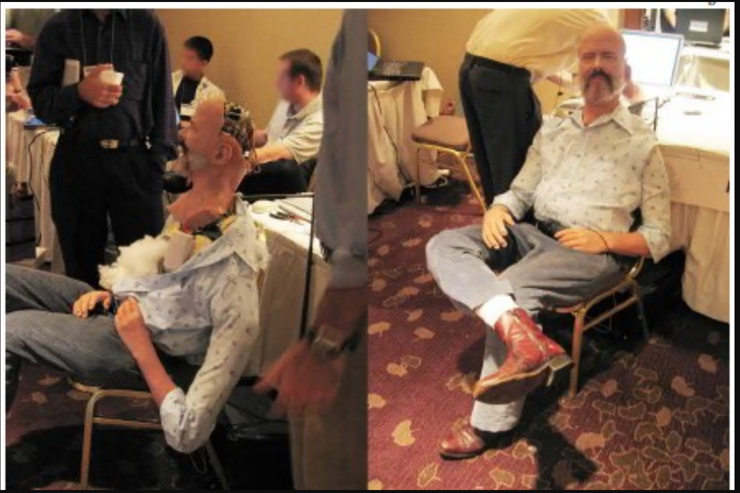
Design Considerations in HRI

- **Design principles**

  3. **Interaction expands function:** When confronted with a robot, people will, in effect, fill in the blanks left open by the design depending on their values, beliefs, needs, and so on. It can thus be useful, particularly for robots with limited capabilities, to design them in a somewhat open-ended way. This allows people to interpret the design in different ways.

  4. **Do not mix metaphors:** Design should be approached holistically- the robot's capabilities, behaviors, affordances for interaction, and so forth should all be coordinated. If you design a humanlike robot, people may find it disturbing if it has skin covering only some parts of its body.

Figure 2.6 Paro (2003–present), a social robot made to resemble a baby harp seal. Paro is provided as a social companion robot. (Source: National Institute of Advanced Industrial Science and Technology)



Anthropomorphization

- Anthropomorphization is the attribution of human traits, emotions, or intentions to nonhuman entities.
- Examples: We all experience anthropomorphism in our daily lives. “My computer hates me!"; “Chuck (the car) is not feeling well lately";
- Pareidolia: The effect of seeing humanlike features in random patterns or mundane objects.

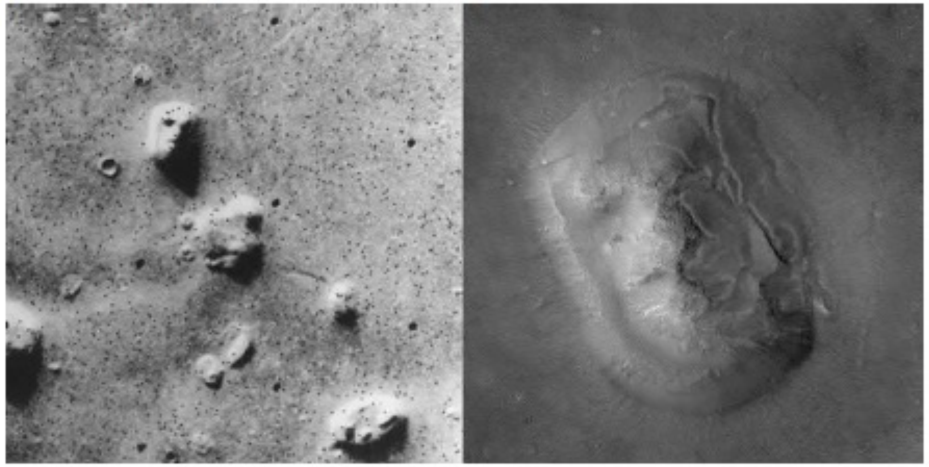
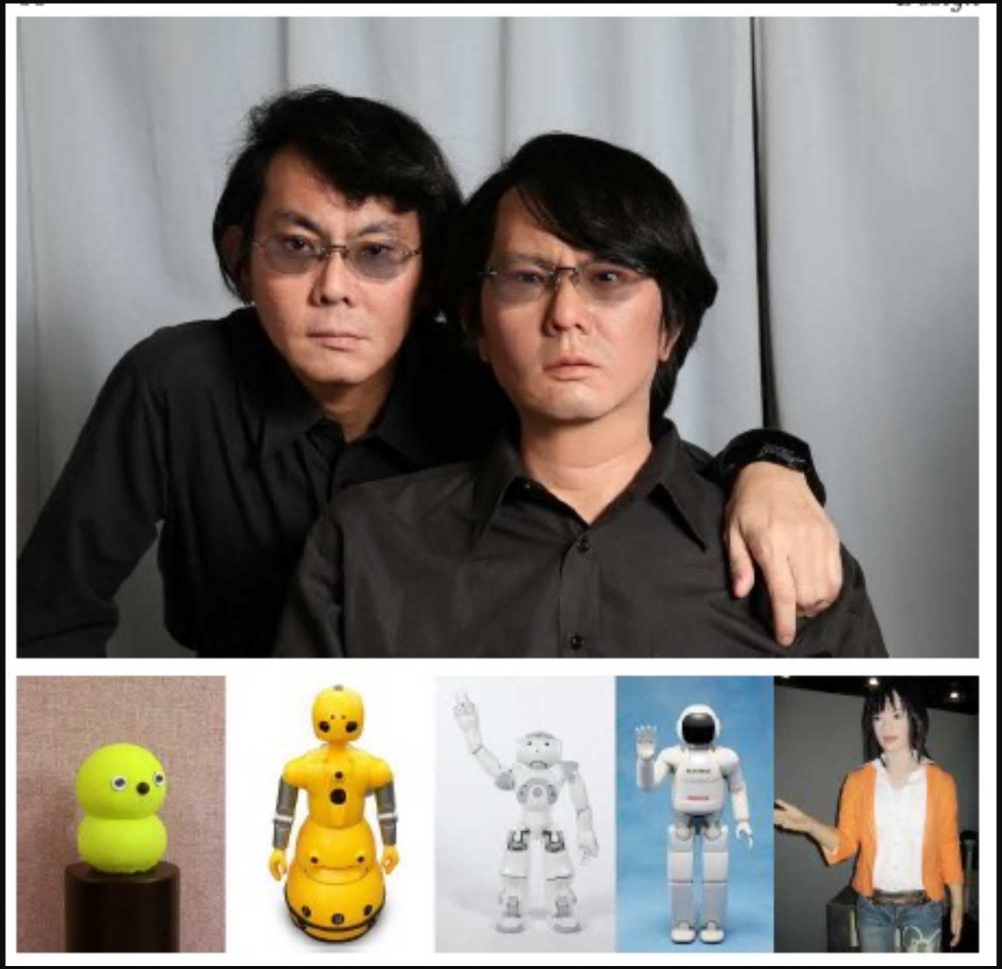


Figure 4.7 The face on Mars is an example of pareidolia. On the left is the photo from 1976, and on the right is the same structure photographed in 2001. (Source: NASA/JPL, NASA/JPL/MSSS)

Anthropomorphization and robots

- In anthropomorphic design, robots are constructed to have certain humanlike characteristics, such as appearance, behavior, or certain social cues, which inspire people to see them as social agents.
- At one extreme, android robots are designed to be as humanlike as possible.
- Humanoid robots, e.g. ASIMO, Nao, etc., use a more abstract notion of human-likeness in their anthropomorphic designs.
- Nonhumanoid robots, e.g. Keepon, can also have anthropomorphic features.
- Anthropomorphism in robot design includes factors related to form and appearance as well as factors relating to behavior.

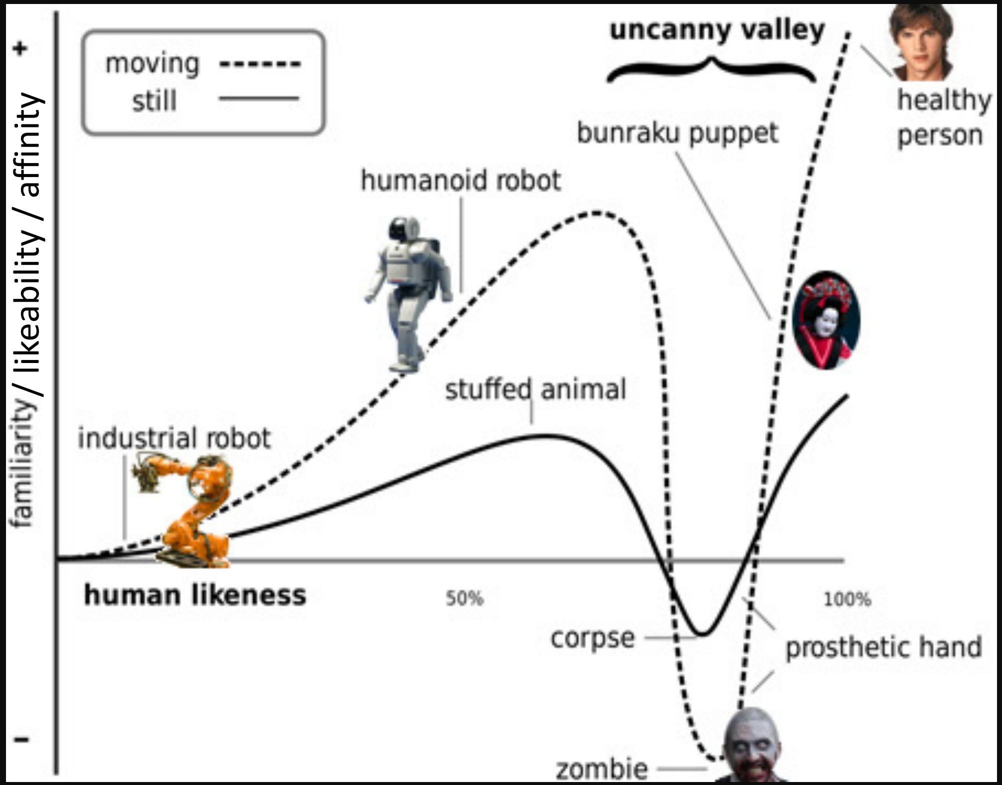


Theorizing anthropomorphism

- Researchers have suggested three core factors that determine anthropomorphic inferences about nonhuman entities:
    1. **Effectance motivation:** It concerns our desire to explain and understand the behavior of others as social actors. This might be activated when people are unsure about how to deal with an unfamiliar interaction partner.
    2. **Sociality motivation:** In this case, people may turn to nonhuman entities as social interaction partners to address their feelings of situational or chronic loneliness.
    3. **Elicited agent knowledge:** It refers to how people use their commonsense understanding of social interactions and actors to understand robots.

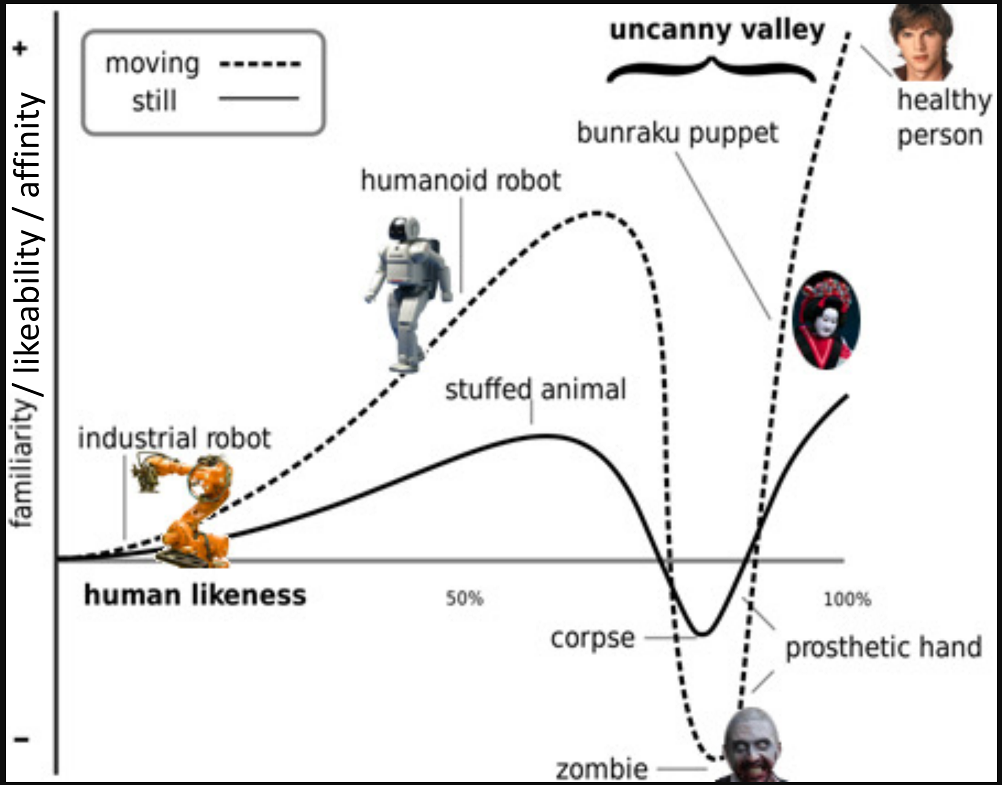
Theorizing anthropomorphism

- **The Uncanny Valley**
  - Mori (1970) made a prediction about the relationship between the anthropomorphism of robots and their likeability.
  - The idea is that the more humanlike robots become, the more likable they will be until a point where they are almost indistinguishable from humans, at which point their likability decreases dramatically.
  - This effect is then amplified by the ability of the robot to move.
  - It is often used to explain why certain robots are being perceived unfavorably, without studying the exact relationship between the features of the robot at hand and its likability.



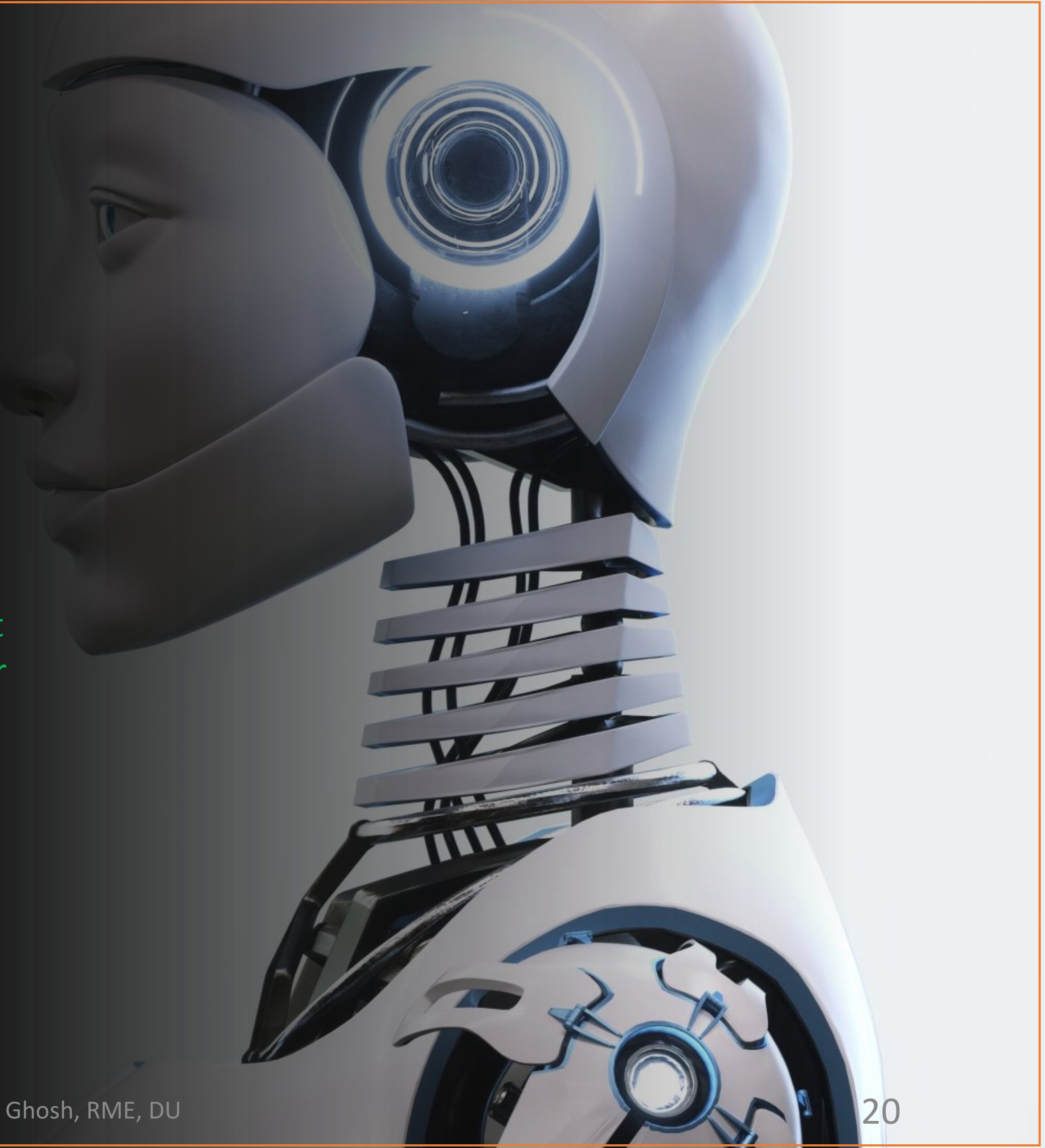
Theorizing anthropomorphism

- **The Uncanny Valley**
  - It is important to note that Mori only proposed this idea and never did any empirical work to test his ideas.
  - A simple possible explanation of why humanlike robots are liked less than, for example, toy robots, is that the difficulty of designing a robot to perform to user expectations increases with its complexity.
  - However, anthropomorphism is a multidimensional concept, and reducing it to just one dimension does not model reality adequately.



Designing anthropomorphism

- To trigger anthropomorphic inferences, robot designers can consider the dimensions of robot appearance and behavior, among many other aspects.
- **Robot appearance**
  - Anthropomorphism in robots can be very simple: just having two dots suggesting eyes and a simple nose or mouth is sufficient to suggest the robot is humanlike.
  - This can be further enhanced by adding more human features, such as arms or legs, but these do not necessarily do very much to further increase the anthropomorphization.
  - Whereas androids mimic human appearance in most ways, simple robots such as Keepon are already very effective at triggering people to anthropomorphize.

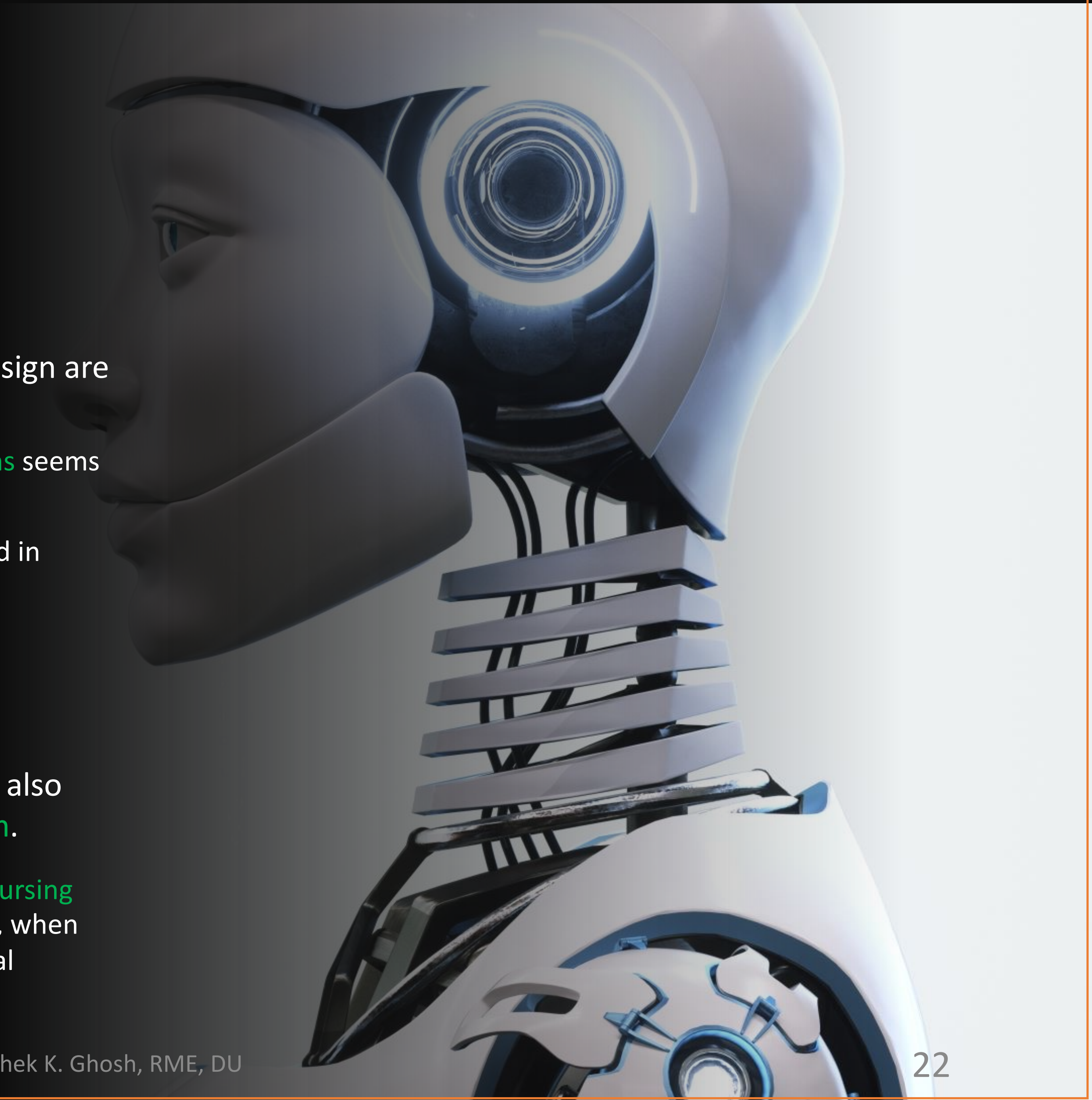


Designing anthropomorphism

- **Robot behavior**
  - A second approach to increasing anthropomorphization is to design the behavior or an artifact such that people perceive humanlike characteristics in its behavior.
  - Motion, rather than form, can be extremely powerful for expressing emotions and intents.
    - Simple geometric shapes, e.g. triangles and circles, moving against a white background evoked people to describe their interactions in terms involving social relationships.
    - A robot vacuum cleaner trying to wriggle its way out from under a table will be described as “being lost" or “not knowing what it wants”.

Designing anthropomorphism

- **Impact of context, culture, and personality**
  - People's perceptions of anthropomorphic robot design are often affected by contextual factors.
    - Just putting a robot in a social situation with humans seems to increase the likelihood of anthropomorphization.
    - The collaborative industrial Baxter robot, when used in factories alongside human workers, was regularly anthropomorphized by them.
  - Some people are more likely than others to anthropomorphize things around them.
  - Seeing other people anthropomorphize robots can also suggest that anthropomorphization is a social norm.
    - Example: Researchers found that older adults in a nursing home were more likely to engage socially with Paro, when they saw others interacting with it like a pet or social companion.



Measuring anthropomorphization

- Researchers also need to know how to measure its presence in an interaction.
- A measure for anthropomorphism specifically developed for HRI is the Godspeed questionnaire.
- It has been widely used in the field and has been translated into several languages.
- More recently, researchers have started developing additional related scales, such as the ROSAS scale and the revised Godspeed questionnaire.
- More subtle behavioral indicators (e.g., language use, application of social norms that are used in human-human interaction) may also be used to investigate the consequences of implementing humanlike form and function in social robots.

GODSPEED I: ANTHROPOMORPHISM

Please rate your impression of the robot on these scales:

以下のスケールに基づいてこのロボットの印象を評価してください。

Fake 偽物のような	1	2	3	4	5	Natural 自然な
Machinelike 機械的	1	2	3	4	5	Humanlike 人間的
Unconscious 意識を持たない	1	2	3	4	5	Conscious 意識を持っている
Artificial 人工的	1	2	3	4	5	Lifelike 生物的
Moving rigidly ぎこちない動き	1	2	3	4	5	Moving elegantly洗練された動き

GODSPEED II: ANIMACY

Please rate your impression of the robot on these scales:

以下のスケールに基づいてこのロボットの印象を評価してください。

Dead 死んでいる	1	2	3	4	5	Alive 生きている
Stagnant 活気のない	1	2	3	4	5	Lively 生き生きとした
Mechanical 機械的な	1	2	3	4	5	Organic 有機的な
Artificial 人工的な	1	2	3	4	5	Lifelike 生物的な
Inert 不活発な	1	2	3	4	5	Interactive 対話的な
Apathetic 無関心な	1	2	3	4	5	Responsive 反応のある

GODSPEED III: LIKEABILITY

Please rate your impression of the robot on these scales:

以下のスケールに基づいてこのロボットの印象を評価してください。

Dislike 嫌い	1	2	3	4	5	Like 好き
Unfriendly 親しみにくい	1	2	3	4	5	Friendly 親しみやすい
Unkind 不親切な	1	2	3	4	5	Kind 親切な
Unpleasant 不愉快な	1	2	3	4	5	Pleasant 愉快な
Awful ひどい	1	2	3	4	5	Nice 良い

GODSPEED IV: PERCEIVED INTELLIGENCE

Please rate your impression of the robot on these scales:

以下のスケールに基づいてこのロボットの印象を評価してください。

Incompetent 無能な	1	2	3	4	5	Competent 有能な
Ignorant 無知な	1	2	3	4	5	Knowledgeable 物知りな
Irresponsible 無責任な	1	2	3	4	5	Responsible 責任のある
Unintelligent 知的でない,	1	2	3	4	5	Intelligent 知的な
Foolish 愚かな	1	2	3	4	5	Sensible 賢明な

GODSPEED V: PERCEIVED SAFETY

Please rate your emotional state on these scales:

以下のスケールに基づいてあなたの心の状態を評価してください。

Anxious 不安な	1	2	3	4	5	Relaxed 落ち着いた
Agitated 動揺している	1	2	3	4	5	Calm 冷静な
Quiescent 平穏な	1	2	3	4	5	Surprised 驚いた

Anthropomorphization

- What is your opinion: Should a social robot have very few humanlike cues, or should it be highly anthropomorphic in design (e.g., like an android)?
- Think about a robot that you might want to have in the near future. Picturing this robot, try to think about a way to encourage more anthropomorphization based on its behavior. Which behaviors should the robot show to be perceived as humanlike?

Design methods

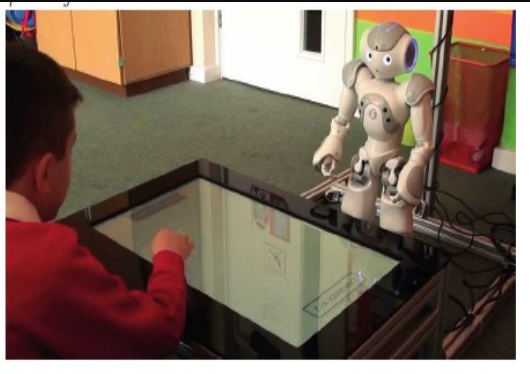
- Design in HRI spans a variety of methods inspired by practice from various disciplines.
- Depending on the method, the starting point and focus of design may weigh more heavily on technical exploration and development or on exploring human needs and preferences.
- But the ultimate goal of design in HRI is to bring these two domains together to construct a successful HRI system.
- The design process is often cyclical in nature, following this pattern:
    1. Define the problem or question.
    2. Build the interaction.
    3. Test.
    4. Analyze.
    5. Repeat from step 2 until satisfied (or money and time run out).

Design methods: Engineering design process

- Commonly used design method in engineering.
- Starting from a problem definition and a set of requirements, numerous possible solutions are considered, and a rational decision is made on which solution best satisfies the requirements.
- Often, the function of an engineered solution can be modeled and then simulated, which allow engineers to systematically manipulate all the design parameters and calculate the resulting properties of the machine
- However, validating a solution in simulation is not always possible. The simulation might not be able to capture the real world in sufficient detail.
- Another factor to consider is to focus not on producing the absolute best solution, but on producing satisficing solutions.
- Satisficing is a portmanteau of satisfy and suffice, meaning that the resulting solution will be just good enough for the purpose it is meant to serve. This is a common problem-solving approach in all human endeavors, and it is almost unavoidable in HRI.

Design methods: Engineering design process

- Some design problems can be ill-defined, or insufficient information is available about the requirements or the environment: “wicked design problem"
- HRI design often is a wicked design problem because there is a lack of information about the appropriate behaviors and consequences of robots in social contexts.
- Example:
  - Engineers working in HRI tried to design a robot to teach eight and nine-year-olds what prime numbers are. They programmed the robot to make eye contact, use the child’s first name, and politely support the child during the quite taxing exercises.
  - They compared the friendly robot against a robot in which the software to maintain an engaging relation was switched off, expecting that robot to be the worse teacher.
  - They were dumbfounded when the aloof robot turned out to be the better teacher by a large margin.
- Relying only on the engineering design method can guide HRI development only so far, particularly when the intended uses of HRI are in open-ended.



Boy learning math with a robot

Design methods: User-centered design process

- User-centered design (UCD) is a broad term used to describe design processes in which end-users influence how a design takes shape.
- The users can be involved in many ways, including
  - through initial analyses of their needs and desires that can help to define the design problem,
  - by asking them to comment on potential robot design variations to see which ones are preferable, and
  - through evaluations of various design iterations of the robot and of the final product to evaluate its success among different users and in different use contexts.

Figure 4.12
Snackbot (2010), a system developed at Carnegie Mellon University to study robots in real-world settings.

Design methods: User-centered design process

- Example
  - Carnegie Mellon University's Snackbot was designed through a user-centered process that involved taking into consideration the robot, people, and the context.
  - It was iteratively performed over 24 months and involved research on
    - where people could already get snacks in the building to establish need,
    - initial technology feasibility and interaction studies,
    - multiple prototypes, and
    - further studies of how the robot was used and the effects of different forms of dialogue and robot behaviors on user satisfaction.

Figure 4.12 Snackbot (2010), a system developed at Carnegie Mellon University to study robots in real-world settings.

Design methods: Participatory design

- Participatory methods seek to include the potential users and other stakeholders, or people who might be affected by robots, in the process of making decisions about appropriate robot design from early on in the design process.
- This is clearly distinct from the notion of bringing users in at the evaluation stage, where the design is partially or fully formed, and users' input is largely used to test particular factors and assumptions already expressed in the design.
- In this way, participatory design recognizes the expertise people have about their everyday experiences and circumstances.
- Example: Roboticists and visually impaired community members and designers worked together in a series of workshops to develop appropriate guidance behaviors for a mobile robot.

Culture in HRI

- Cultural considerations should be taken into account when designing robots both for international and local users.
- Researchers commonly make connections between cultural traditions and the design and use of robots, particularly contrasting the norms, values, and beliefs in the East and West:
    - Animist beliefs have been used to explain the perceived comfort of Japanese and Korean populations with robots (Geraci, 2006; Kaplan, 2004; Kitano, 2006),
    - whereas human exceptionalism has been suggested as a source of Westerners' discomfort with social and humanoid robots (Geraci, 2006; Brooks, 2003).
- HRI researchers have been studying how cultural differences affects people's perceptions of face-to-face encounters with robots.
    - In a comparison using Dutch, Chinese, German, U.S., Japanese, and Mexican participants, it was found that U.S. participants were the least negative toward robots, whereas the Mexican participants were the most negative.
    - Against expectations, the Japanese participants did not have a particularly positive attitude toward robots (Bartneck et al., 2005).

Planning human studies

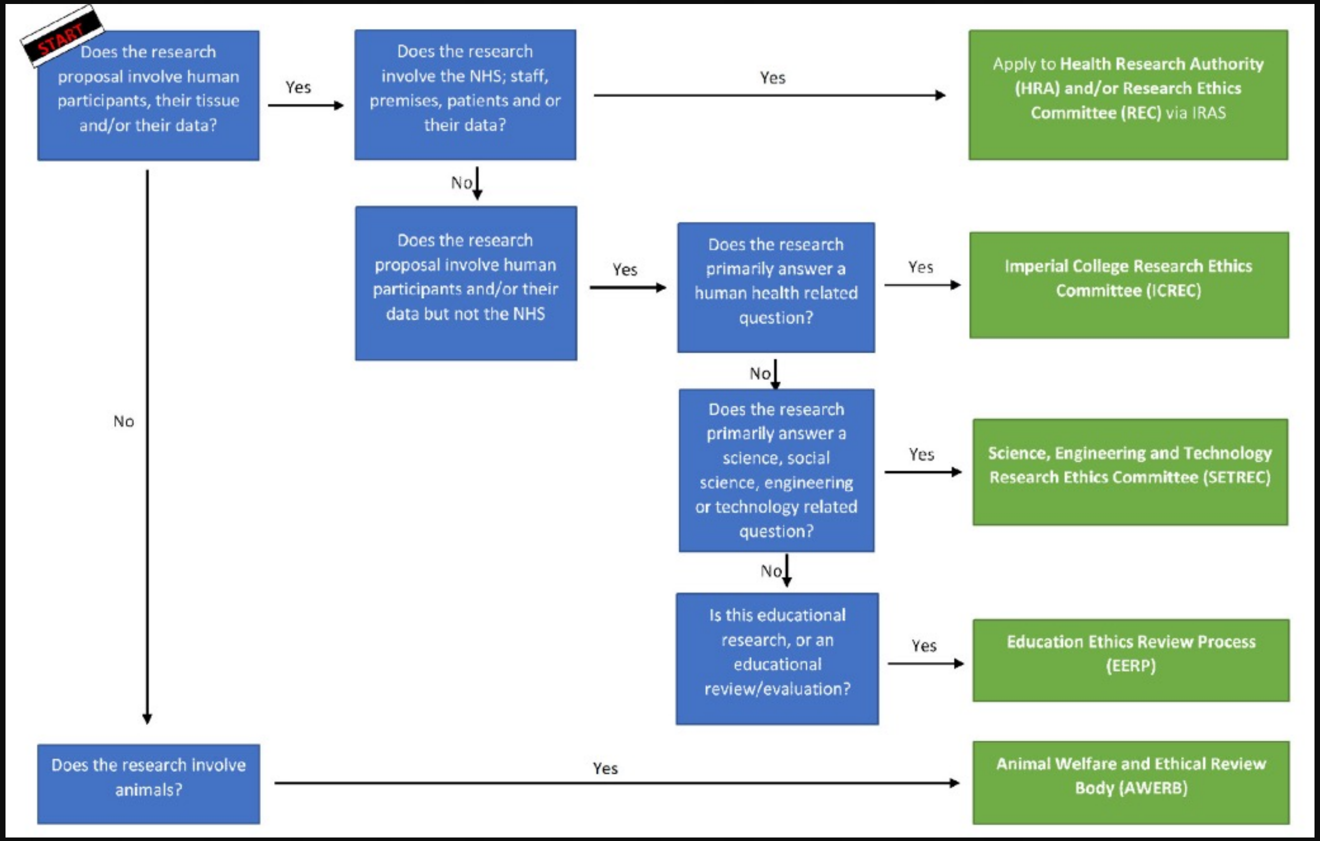
- **Ethical approval for research work: Why needed?**
  - Ethics reviews mean ensuring that a research project is conducted in such a way that any **potential adverse impact is in proportion to its expected benefit.**
  - **Adverse impacts** are those that may affect the environment and/or the health, well-being, safety, threats to privacy and in rare cases liberty of individuals and/or the reputation of the College.
  - **Adverse impacts** may consist of **acknowledged potential direct risks** (such as rare or adverse reaction to researcher intervention), **unintended consequences** (such as stigmatization of an individual through involvement in research).

Planning human studies

- **Ethical approval for research work: What to consider?**
  - Does the work has the potential to damage the mental or physical health of human participants, (e.g. volunteers, College staff and students, or patients,) or others who may be affected?
  - Does the work has the potential to jeopardise the safety and liberty of people affected by the research (e.g. volunteers working in sensitive situations or abroad)?
  - Does the work has the potential to compromise the privacy of individuals whose data is involved in the work?
  - Does the work involves methods (e.g. genetic research, interviews, questionnaires, randomised control trial) or subject matter (e.g. recreational and controlled drugs, human impact on the environment) that are sensitive and therefore need to be managed consistently with the College's high public reputation?
  - Does the work carries a risk of an actual or perceived conflict of interest on the part of researchers and/or the College?
  - Does the work has a potential dual purpose in addition to the intended purpose?

Case study: Ethical approval process for Imperial College London, UK

- Who needs to apply?
  - Ethics approval is needed for any research that involves human participants; their tissue and /or data to ensure that the dignity, rights, safety and well-being of all participants are the primary consideration of the research project.



Case study: Ethical approval process for Imperial College London, UK

- Preparing the application
    - Fill up all the necessary forms

Ethics Application Checklist for Imperial College Research Ethics Committee (ICREC) and Science, Engineering and Technology Research Ethics Committee (SETREC)

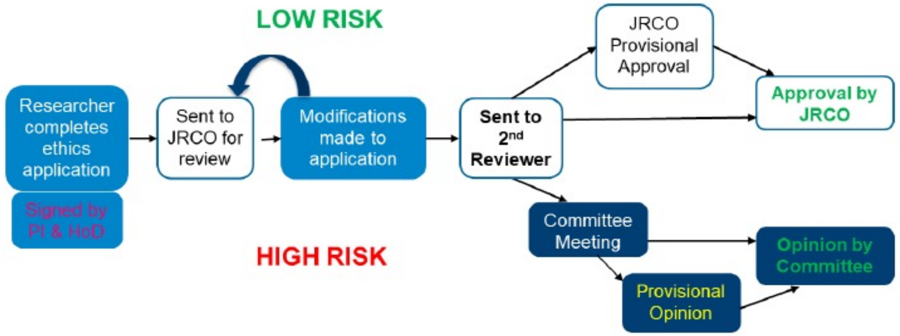
Documents	ICREC Primary Data Collection	ICREC Secondary Data	SETREC Primary Data Collection	SEREC Secondary Data
1. ICREC-SETREC Application Form (signed by PI and HoD) (M)				
2. Protocol (M)				
3. JRCO Sponsorship and Insurance Request Form (M)				
4. Participant Information Sheet				
5. Consent Form				
6. Recruitment adverts				
7. Recruitment sample email				
8. Questionnaires, interview questions, focus group topic guide				
9. Other documents (e.g. local ethics approval)				
10. Relevant appendices				
a. Appendix I – Research involving ionizing radioactive substances/x-rays				
b. Appendix II – Research involving the use of genetically modified materials (attach letter from GM Safety Committee)				
a. If conducting research using drugs/medication – SmPc attached				

(M) Mandatory documents for ethics approval

Case study: Ethical approval process for Imperial College London, UK

- Apply for the ethics

Appendix 2: Ethics process flow diagram.



JRCO = Joint research compliance office

	1 Low risk	2 High risk
	If you answer yes to all the questions in column 1 your research is considered low risk.	If you answer yes to at least one of the questions in column 2 your research is considered high risk.
Participants	• participants are all 16 years and over. • the participants are not considered vulnerable.	• Vulnerable groups such as children, adults who are unable to consent or individuals with learning difficulties • prisoners or young offenders? • without their consent
Nature of the study	• no risk to the participant that might lead to disclosures of a sensitive nature or cause anxiety	• an element of deception • disclosure of illegal or harmful activity or sensitive information • induces stress, anxiety or negative consequences • where the researcher or gatekeepers are in a position of influence or authority over participants that could give rise to a perceived pressure to participate
Study methodology	• there no drugs/ placebos involved • data has already been collected or is pre-existing data • data from people speaking in public places • no invasive techniques or use of human tissue	• using ultrasound or sources of non-ionising radiation • physically intrusive involving administration of substances, use of bodily materials, or DNA/RNA analysis. • medical devices that are CE marked used outside its product limitation or any non-CE marked products
Geographical Location	• UK • online surveys UK and abroad	• areas of political unrest • areas with public health concerns • local ethics approval needed
Other reasons		• undue incentives are offered

Case study: Ethical approval process for University of Dhaka

- Different faculties have their ethical committee.
- For our department, we need to apply to the Dean of the Faculty of Engineering and Technology for our ethical approval.
- Documents to be filled up are-
  - Application form
  - Participant information sheet
  - Participant’s information form
  - Participants’ consent form
  - Interview/Survey questions

Assignment:

1. Team up with one other student
2. Plan an HRI study that you may want to perform as your research work
3. Prepare an ethical application for your study following the guidelines from the Faculty of Engineering and Technology, University of Dhaka, and fill up the following documents -
   i. Application form
   ii. Participant information sheet
   iii. Participant's information form
   iv. Participants' consent form
   v. Interview/Survey questions

The End